Exercise C1

Q1. If one root of the quadratic equation $ax_2 + bx + c = 0$ be nth root of the other then $(acn)^{1/n+1} + (anc)^{1/n+1} =$

(A) $b^{1/n+1}$ (B) $b^{1/n}$ (C) b (D) $-b$

Ans. D

Sol. $\alpha . \alpha^n = \frac{c}{a}$ $\Rightarrow$ $\alpha = \left(\frac{c}{a}\right)^{\frac{1}{n+1}}$ $\Rightarrow \alpha + \alpha^n = \frac{b}{a} \Rightarrow$ $\left(\frac{c}{a}\right)^{\frac{n}{n+1}} = -\frac{b}{a}$

$a^{1-\frac{1}{n+1}} . c^{\frac{1}{n+1}} + c^{\frac{n}{n+1}} . a^{1-\frac{n}{b+1}} + b = 0$ $\Rightarrow$ $a^{\frac{n}{n+1}} . c^{\frac{1}{n+1}} + a^{\frac{1}{n+1}} . c^{\frac{n}{n+1}} + b = 0$

$\left(a^n c\right)^{\frac{1}{n+1}} + \left(ac^n\right)^{\frac{1}{n+1}} + b = 0$ Proved

Q2. If α, β are the roots of the equation $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$, then a S_{n+1}+c S_{n-1} =

(A) $b\,S_n$ (B) b^2S_n (C) $2bS_n$ (D) $-bS_n$

Ans. D

Sol. -

Q3. If the equation $x^2 + 2\,(k+1)\,x + 9k - 5 = 0$ has only negative roots, then-

(A) $k \le 0$ (B) $k \ge 0$ (C) $k \ge 6$ (D) $k \le 6$

Ans. C

Sol. -

Q4. The set of values of 'a' for which $f(x) = ax^2 + 2x\,(1 - a) - 4$ is negative for exactly three integral values of x, is-

(A) $(0, 2)$ (B) $(0, 1]$ (C) $[1, 2)$ (D) $[2, \infty)$

Ans. C

Sol. -

Q5. The number of real solutions of the equation $(15 + 4\sqrt{14})^t + (15 - 4\sqrt{14})^t = 30$ are, where $t = x^2 - 2|x|$.

(A) 0 (B) 2 (C) 4 (D) 6

Ans. C

Sol. $\Rightarrow x^2 - 2|x| = 1$ but $|x| \neq 1 - \sqrt{2}$

$\Rightarrow x = (1 + \sqrt{2}), 1$ 4 solution

Q6. If α, β are the roots of $ax^2 + bx + c = 0$, $(a \neq 0)$, and $\alpha + \delta, \beta + \delta$ are the roots of $Ax^2 + Bx + C = 0$, $(A \neq 0)$ for some constant δ, then the value of $\frac{b^2 - 4ac}{a^2}$ is

(A) $\frac{B^2 - 4AC}{A^2}$ (B) $\frac{B^2 + 4AC}{A^2}$ (C) $\frac{A^2 - 4BC}{A^2}$ (D) None of these

Ans. A

Sol.

$$ax^2 + bx + c = 0 \begin{matrix} \nearrow \alpha \\ \searrow \beta \end{matrix} \quad \Rightarrow \quad \alpha + \beta = \frac{b}{a}, \alpha\beta = \frac{c}{a} \quad \Rightarrow \quad Ax^2 + Bx + C = r \begin{matrix} \nearrow \alpha + \delta \\ \searrow \beta - \delta \end{matrix}$$

$$(\alpha + = delta)(\beta + \delta) = -\frac{B}{A}, (\alpha + \delta)(\beta + \delta) = \frac{C}{A} \quad \because \ |(\alpha + \delta) - (\beta + \delta)| = |(\alpha - \beta)|$$

$$\Rightarrow \sqrt{\frac{B^2}{A^2} - \frac{4C}{A}} = \sqrt{\frac{b^2}{a^2} - \frac{4c}{a}} \Rightarrow \frac{b^2 - 4ac}{a^2} = \frac{B^2 - 4AC}{A^2}$$

Hence proved

Q7. Given a, b, c are non negative real numbers and if $a^2 + b^2 + c^2 = 1$, then the value of $a + b + c$ is :

(A) ≥ 3 (B) ≥ 2 (C) $\leq \sqrt{2}$ (D) $\leq \sqrt{3}$

Ans. D

Sol. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2\sum ab = 1 + 2\sum ab]$

put $a^2 + b^2 + c^2 \geq ab + bc + ca$

or $\sum ab \leq 1$

$\therefore \ (a + b + c)^2 \leq 1 + 2$

$\therefore (a + b + c)^2 \leq \sqrt{3}$

Alternatively : using RMS $\geq$ AM in a, b, c

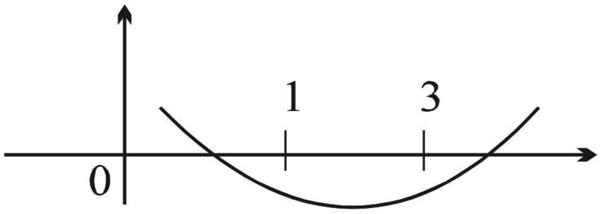
$$\sqrt{\frac{a^2 + b^2 + c^2}{3}} \geq \frac{a + b + c}{3} \Rightarrow \sqrt{3} \geq (a + b + c)^2 \quad \Rightarrow \quad a + b + c \leq \sqrt{3}$$

Q8. The set of values of 'a' for which the inequality, $(x - 3a)(x - a - 3) < 0$ is satisfied for all $x \in [1, 3]$ is:

(A) (1/3, 3) (B) (0, 1/3) (C) (– 2, 0) (D) (– 2, 3)

Ans. B

Sol. Equation is $x^2 - (4a + 3)x + 3a(a + 3)$



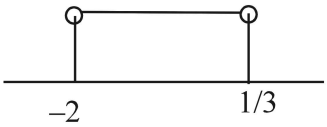
f(1) < 0 and f(3) < 0

(1–3a) (1 – a – 3) < 0 1 – a – 3 – 3a + 3a2 + 9a < 0

$\Rightarrow 3a^2 + 5a - 2 < 0$

$3a^2 + 6a - a - 2 < 0$

$3a(a + 2) - (a + 2) < 0$



$(a + 2)(3a - 1) < 0 \Rightarrow$

again $(3 - 3a)(-a) < 0$ $(a - 1)a < 0$

$\Rightarrow 0 < a < 1$

Hence $a \in \left(0, \frac{1}{3}\right)$

Q9. If α and β are the roots of $x^2 + px + q = 0$ and α^4, β^4 are the roots of $x^2 - rx + s = 0$, then the equation

$x^2 - 4qx + 2q^2 - r = 0$ has always (p, q, r, s $\in$ R) :

(A) two real roots (B) two positive roots (C) two negative roots

(D) one positive and one negative root.

Ans. A

Sol. $\alpha + \beta = -p$; $\alpha\beta = q$; $\alpha^4 + \beta^4 = r =$ r ; $\alpha^4\beta^4 = s$

Now $D = 16q^2 - 4(2q^2 - r) = 8q^2 + 4r = 4(2q^2 + r)$

$= 4[2\alpha^2 b^2 + \alpha^4 + \alpha^4] = 2[(\alpha^2 + \beta^2)^2]$

$= 4(p^2 - 2q)^2] > 0$

Again consider the product of the roots

$= 2q^2 - r$ which can be either positive or negative. Hence A.

Q10. Number of integral values of 'a' for which the equation $x^2 - (a + 1)x + a - 1 = 0$, has integral roots, is equal to

(A) 1 (B) 2 (C) 3 (D) 4

Ans. A

Sol. -

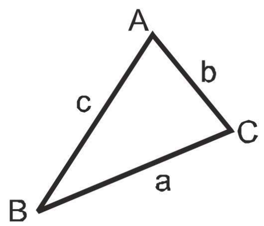
Q11. Let a, b, c be the sides of a triangle. No two of them are equal and $\lambda \in R$. If the roots of the equation
$x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ are real, then

(A) $\lambda < \frac{4}{3}$ (B) $\lambda > \frac{5}{3}$ (C) $\lambda \in \left(\frac{1}{3}\frac{5}{3}\right)$

Ans. A

Sol. We, know that $a + b > c$, $b + c > a$ and $c + a > b$ $\Rightarrow$ $c - a < b$, $a - b < c$, $b - c < a$

squaring on both sides and adding $(c - a)^2 + (a - b)^2 + (b - c)^2 < a^2 + b^2 + c^2$
$a^2 + b^2 + c^2 - 2(ab + bc + ca) < 0 \Rightarrow (a + b + c)^2 - 4(ab + bc + ca) < 0$



$\Rightarrow \quad \frac{(a+b+c)^2}{ab+bc+ca} < 4 \quad$ ..... (i)

Now roots of equation $x^2 + 2(a + b + c) x + 3\lambda (ab + bc + ca) = 0$ are real, then $D \geq 0$

$\Rightarrow \quad 4(a+b+c)^2 - 4.3\lambda.(ab+bc+ca) \leq 0 \quad \Rightarrow \quad \frac{(a+b+c)^2}{ab+bc+ca} \geq 3\lambda$

So $\quad 3\lambda \leq \frac{(a+b+c)^2}{ab+bc+ca} < 4 \quad \Rightarrow \quad \lambda < \frac{4}{3}$

Q12. If α, β are the roots of equation $x^2 + x - 2 = 0$, then the value of
$\frac{\alpha\beta^4(\beta+1)^4 + \beta\alpha^4(\alpha+1)^4}{\alpha^2+\beta^2+\alpha+\beta}$ is equal to

(A) 2 (B) –2 (C) 4 (D) –4

Ans. D

Sol. -

Q13. If α, β, γ are the roots of the cubic $x^3 - 2x + 3 = 0$ then the value of
$\frac{1}{\alpha^3+\beta^3+6} + \frac{1}{\beta^3+\gamma^3+6} + \frac{1}{\gamma^3+\alpha^3+6}$
equals

(A) $\frac{1}{3}$ (B) $\frac{-1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{-1}{2}$

Ans. B

Sol. -

Q14. If α, β be roots of $x^2 + px + 1 = 0$ and γ, δ are the roots of $x^2 + qx + 1 = 0$ then $(\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta) =$

(A) $p^2 + q^2$ (B) $p^2 - q^2$ (C) $q^2 - p^2$ (D) None of these

Ans. C

Sol. -

Q15. If a, b, c are real numbers satisfying the condition $a + b + c = 0$ then the roots of the quadratic equation
$3ax^2 + 5bx + 7c = 0$ are :

(A) positive (B) negative (C) real and distinct (D) imaginary

Ans. C

Sol. $D = 25b^2 - 84\,ac$

$= 25(a + c)^2 - 84ac$ using $b = -(a + c)$

$= 2[(a + c)^2 - 4ac] + 4(a + c)^2 > 0$

Q16. Let a, b, c be three real numbers such that $a + b + c = 0$ and $a^2 + b^2 + c^2 = 2$. Then the value of $(a^4 + b^4 + c^4)$ is equal to

(A) 2 (B) 5 (C) 6 (D) 8

Ans. A

Sol. Given $a + b + c = 0$

and $a^2 + b^2 + c^2 = 2$

$(a^2 + b^2 + c^2)^2 = 4$

$a^4 + b^4 + c^4 + 2[a^2b^2 + b^2c^2 + c^2a^2] = 4$

let $a^4 + b^4 + c^4 = E$

hence $E + 2[(ab + bc + ca)^2 - 2abc(a + b + c)] = 4$

$\therefore$ $E + 2(ab + bc + ca)^2 = 4$ (as $a + b + c = 0$)

again, $(a + b + c)^2 = 0$ $\Rightarrow \sum a^2 + 2\sum ab = 0$

$2 + 2(ab + bc + ca) = 0$

$\therefore$ $ab + bc + ca = -1$

$\therefore$ $E + 2 = 4$; $\therefore$ $E = 2$ **Ans.**

Q17. The number of solution of the equation $e^{2x} + e^x + e^{-2x} + e^{-x} = 3(e^{-2x} + e^x)$ is

(A) 0 (B) 2 (C) 1 (D) more than 2

Ans. C

Sol. $x = \ln 2$

Q18. Complete set of values of 'a' such that $y = \dfrac{x^2 - x}{1 - ax}$ (x ∈ R) attains all real values is :

(A) [1, 4] (B) (0, 4] (C) (0, 1] (D) (1, ∞)

Ans. D

Sol. $y = \dfrac{x^2 - x}{1 - ax}$

$$y = \frac{x(x-1)}{(1-ax)}, a \neq 1$$

$y - axy = x^2 - x$

$x^2 + x - axy - y = 0$

$x^2 + (1 - ay)x - y = 0$

$D \geq 0$

$(1 - ay)^2 + 4y \geq 0$

$1 + a^2y^2 - 2ay + 4y \geq 0$

$a^2y^2 + (4 - 2a)y + 1 \geq 0$

$D \leq 0$

$(4 - 2a)^2 - 4a^2 \leq 0$

$16 + 4a^2 - 16a - 4a^2 \leq 0$

$16(a - 1) \geq 0$

$a \geq 1$

$a \in (1, \infty)$ $\quad \therefore a \neq 1$

Q19. f the roots of the equation $bx^2 + cx + a = 0$ be imaginary, then for all real values of x, the expression $3b^2x^2 + 6bcx + 2c^2$ is -

(A) Greater than 4 ab (B) Less than 4ab (C) Greater than –4ab

(D) Less than – 4ab

Ans. C

Sol. $bx^2 + cx + a = 0$

Roots are imaginary $c^2 - 4ab < 0$

$f(x) = 3b^2x^2 + 6bcx + 2c^2$

$D = 36b^2c^2 - 24b^2c^2 = 12b^2c^2$

$\because 3b^2 > 0$

$\therefore f(x) \geq \left(-\frac{D}{4a}\right)$

$f x ( ) \geq -c^2$

Now $c^2 - 4ab < 0$

$c^2 < 4ab$

$-c^2 > -4ab$

$\therefore f(x) > -4ab$

Q20. Let for $a \neq a_1 \neq 0$, $f(x) = ax^2 + bx + c$, $g(x) = a_1x^2 + b_1x + c_1$ and $p(x) = f(x) - g(x)$. If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then the value of $p(2)$ is :

(A) 18 (B) 3 (C) 9 (D) 6

Ans. A

Sol. $P(x) = 0 \Rightarrow f(x) = g(x) \Rightarrow ax^2 + bx + c = a_1x^2 + b_1x + C,$

$\Rightarrow (a - a_1)x^2 + (b - b_1)x + (c - c_1) = 0.$

It has only one solution $x = -1$

$\Rightarrow b - b_1 = a - a_1 + c - c_1$ __________(1)

vertex $(-1, 0)$ $\Rightarrow \dfrac{b - b_1}{2(a - a_1)} = -1 \Rightarrow b - b_1 = 2(a - a_1)$ __________(2)

$\Rightarrow f(-2) - g(-2) = 2 \Rightarrow 4a - 2b + c - 4a_1 + 2b_1 - c_1 = 2$

$\Rightarrow 4(a - a_1) - 2(b - b_1) + (c - c_1) = 2$ .... (3)

by (1), (2) and (3) $(a - a_1) = (c - c_1) = \frac{1}{2}(b - b_1) = 2$

Now $P(2) = f(2) - g(2) = 4(a - a_1) + 2(b - b_1) + (c - c_1) = 8 + 8 + 2 = 18$

Q21. The equation $e^{sin\,x} - e^{-sin\,x} - 4 = 0$ has

(A) no real roots (B) exactly one real root (C) exactly four real roots

(D) infinite number of real roots

Ans. A

Sol. Given, $e^{\sin x} - e^{-\sin x} - 4 = 0$

Let $e^{\sin x} = y$. Then the given equation becomes

$y - \frac{1}{y} = 4 \Rightarrow y^2 - 4y - 1 = 0$

$\Rightarrow y = \frac{4 \pm \sqrt{16+4}}{2} = 2 \pm \sqrt{5}$

$\Rightarrow e^{\sin x} = 2 \pm \sqrt{5} \Rightarrow \sin x = \log_e \left(2 \pm \sqrt{5}\right)$

$\Rightarrow \sin x = \log_e \left(2 + \sqrt{5}\right)$

$(\because \left(2 - \sqrt{5}\right) < 0$ and so $\log_e \left(2 - \sqrt{5}\right) > 1$is not defined)

But the value of sin x lies between – 1 and 1, both values inclusive, so

$\sin x \neq \log_e \left(2 + \sqrt{5}\right)$

$\therefore$ There are not possible real roots of the given equation.

Q22. If α and β are roots of the equation $px^2 + qx + r = 0$, $p \neq 0$. If p, q , r are in AP and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$ then value of $|\alpha - \beta|$ is

(A) $\frac{\sqrt{34}}{9}$ (B) $\frac{2\sqrt{13}}{9}$ (C) $\frac{\sqrt{61}}{9}$ (D) $\frac{2\sqrt{17}}{9}$

Ans. B

Sol.

$px^2 + qx + r = 0 \begin{array}{l} \nearrow \alpha \\ \searrow \beta \end{array}$; p, q, r $\to$ A.P. ; 2q = p + r

$\frac{1}{\alpha} + \frac{1}{\beta} = 4$; $\frac{\alpha + \beta}{\alpha\beta} = 4 \Rightarrow \frac{-q}{r} = 4$

q = – 4r ..... (i)

$\therefore$ –8r = p + r

p = – 9r ..... (ii)

$|\alpha - \beta| = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} = \sqrt{\frac{q^2}{p^2} - \frac{4r}{p}}$ by (i) and (ii)

$= \frac{\sqrt{q^2 - 4qr}}{|p|} = \frac{\sqrt{16r^2 + 36r^2}}{|-9r|} = \frac{2\sqrt{13}}{9}$

Q23. Which of the following statement(s) is/are True?

(A) The equation $\sqrt{x+1} + \sqrt{x-1} = 1$ has no real solution.

(B) If $0 < p < \pi$ then the quadratic equation, $(\cos p - 1)\, x^2 + \cos px + \sin p = 0$ has real roots.

(C) If $2a + b + c = 0$ $(c \neq 0)$ then the quadratic equation, $ax^2 + bx + c = 0$ has no root in (0, 2).

(D) If x and y are positive real numbers and m , n are any positive integers then ; $\frac{x^n . y^m}{(1 + x^{2n})(1 + y^{2m})} > \frac{1}{4}$.

Ans. A, B

Sol. (A) $x = \frac{5}{4}$ is rejected

(C) note that f(0) and f(2) have opposing signs under the given condition.

Q24. Let $(a - 1)(x^2 + \sqrt{3}x + 1)^2 - (a + 1)(x^4 - x^2 + 1) \leq 0 \ \forall x \in R$, then which of the following is/are correct?

(A) $a \in \left[-\frac{1}{\sqrt{3}}, \frac{4}{\sqrt{3}}\right]$ (B) Largest possible value of a is $\sqrt{3}$

(C) Number of possible integral values of a is 3

(D) Sum of all possible integral values of a is '0'

Ans. D, C

Sol. -

Q25. Let $f(x) = x^2 + ax + b$. If the maximum and the minimum values of $f(x)$ are 3 and 2 respectively for $0 \leq x \leq 2$, then the possible ordered pair(s) of (a, b) is/are

(A) $(-2, 3)$ (B) $(-3/2, 2)$ (C) $(-5/2, 3)$ (D) $(-5/2, 2)$

Ans. A

Sol. -

Q26. If the expression $y = \frac{x - k}{x^2 - 4x + k}$ can take all real values for permissible real values of x, then the possible integral value of k can be

(A) 0 (B) 1 (C) 2 (D) 3

Ans. B, C

Sol. -